Stage 1: Topic Selection and Initial Proposal Plan

Proposed Title

Human-in-the-Loop AI in Accounting Automation: Managing Errors, Uncertainty, and Accountability in Bookkeeping Decisions

Background of the Study

Artificial intelligence is now widely used in accounting and bookkeeping to handle routine financial tasks. After reviewing ten academic article abstracts, it appears that AI facilitates invoice processing, transaction classification, account reconciliation, financial reporting, auditing, fraud monitoring, error detection, and decision-making. The reviewed studies show that AI reduces manual work, boosts efficiency, speeds up financial data processing, and lets accounting professionals spend more time on review, analysis, and quality assurance.

Even with these benefits, using AI for accounting automation comes with important challenges. Sometimes, AI systems give results that are wrong, unclear, or hard to understand, especially if the financial data is missing, confusing, or unusual. The abstracts reviewed suggest that while AI can make accounting better and cut down on certain mistakes, how well it works depends on how the system is built, how closely it is monitored, and how accounting professionals use it.

Human-in-the-loop AI is important because it keeps humans involved in automated decision-making. Instead of allowing AI systems to make final decisions alone, human-in-the-loop systems allow users to review, correct, approve, or override AI outputs. Therefore, this topic matters because accounting automation is about more than speed and efficiency. It also involves accuracy, error management, uncertainty, ethical responsibility, and accountability in bookkeeping decisions.

Problem Statement

Artificial intelligence (AI) is increasingly utilized to automate accounting and bookkeeping processes. Numerous studies highlight its advantages, including accelerated processing, reduced manual intervention, enhanced efficiency, fewer errors, and improved support for accounting and auditing functions.

However, optimal accounting automation extends beyond increased processing speed. An effective system should demonstrate efficiency, accuracy, transparency, and accountability. In automated bookkeeping, AI-generated entries must be reliable, uncertain outputs require human review, errors should be identified, and ultimate responsibility must be clearly defined.

The problem is that AI accounting systems may focus heavily on automation but may not always clearly explain when human review is needed, how to handle uncertain AI outputs, or who is responsible when AI-generated bookkeeping decisions are wrong. This research aims to address this problem by examining how human-in-the-loop AI can support better error management, uncertainty handling, and accountability in AI-assisted bookkeeping decisions.

Research Gap

Prior research has investigated AI adoption in accounting, the implementation of AI-powered accounting systems, automated accounting processes, the application of AI in auditing, and the impact of AI on reducing accounting errors. These studies explain how AI can support automation and enhance accounting efficiency.

However, previous research has not fully examined how human review should be designed within AI-assisted bookkeeping systems. In particular, there is still limited understanding of when accountants should intervene, how uncertain AI-generated entries should be flagged, how AI errors should be corrected, and how accountability should be maintained when bookkeeping decisions are partly AI-generated.

Although human-in-the-loop AI has been studied more broadly, there remains a need to connect this concept specifically to bookkeeping decisions in accounting automation.

Aim

This study aims to examine how human-in-the-loop AI can help manage errors, uncertainty, and accountability in automated bookkeeping decisions.

Objectives

1. To review academic literature on AI automation in accounting and bookkeeping.
2. To examine the main risks of AI-generated bookkeeping decisions, including errors, uncertainty, and unclear accountability.
3. To explore how human-in-the-loop AI can support the review, correction, and approval of AI-generated accounting entries.
4. To develop a conceptual understanding of how human oversight can make AI-assisted bookkeeping more reliable, responsible, and accountable.

Research Questions

1. How can human-in-the-loop AI improve the reliability of automated bookkeeping decisions?
2. What types of errors and uncertainty may occur in AI-generated accounting entries?
3. How should human accountants review, correct, or approve AI-generated bookkeeping outputs?
4. How can accountability be maintained when bookkeeping decisions are supported by AI systems?